

Patient-Specific Computational Fluid Dynamics of the Airways

Segmentation, mesh generation, and flow analysis in tracheomalacia

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Computational Biofluid Mechanics

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`github.com/henrivanovermeire/tracheomalacia_cfd`

Clinical and computational question

- Tracheomalacia and congenital stenosis can strongly increase airway resistance.
- CT describes anatomy; CFD estimates velocity, pressure loss, resistance, and flow division.
- Preoperative and postoperative anatomy were reconstructed at corresponding tracheal locations.

Main question

How does the intervention alter lumen calibre and the geometry-driven airflow through the reconstructed airway?

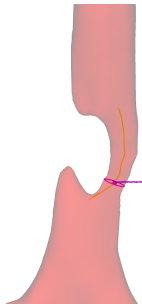


Image to result

DICOM → 3D Slicer/SlicerVMTK → capped STL → Gmsh → OpenFOAM → ParaView

Geometry preparation

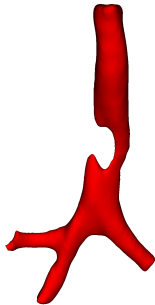
- Case-specific airway segmentation
- Centerline-normal measurements
- Centerline-directed extensions and planar caps

Flow solution

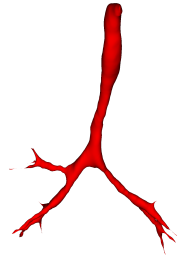
- Tetrahedral HXT meshes
- Steady laminar simpleFoam
- Transient postoperative pimpleFoam PoC

Patient-specific airway reconstruction

Preoperative



Postoperative



The preoperative stenosis was preserved during segmentation and manually separated from included lung air.

Intervention increased the matched lumen area

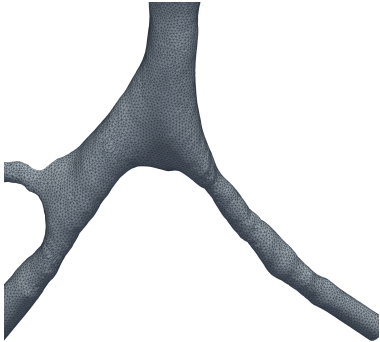


Quantity	Preoperative	Postoperative
Area (mm ²)	1.813	5.169
Minimum Feret diameter (mm)	0.798	2.292
Equivalent diameter (mm)	1.519	2.565

Area constriction: 64.9%; minimum-Feret reduction: 65.2%.

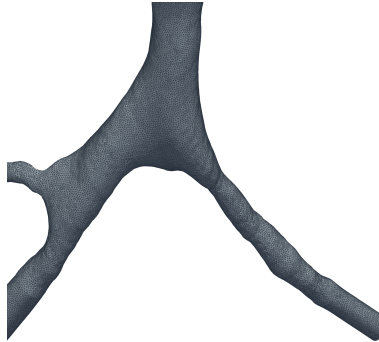
Mesh generation and verification

0.25 mm baseline



180,163 cells

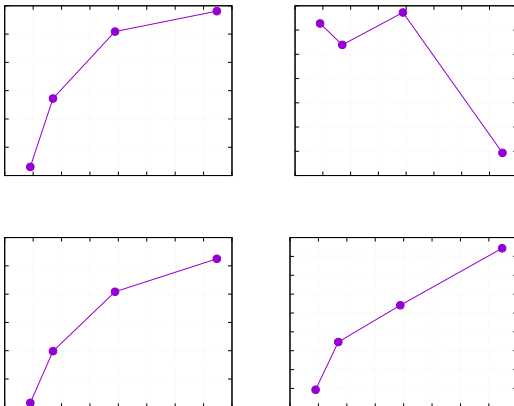
0.15 mm selected



776,568 cells

The 0.15 mm HXT mesh had no faces above 70° non-orthogonality and provided the best accuracy–cost balance.

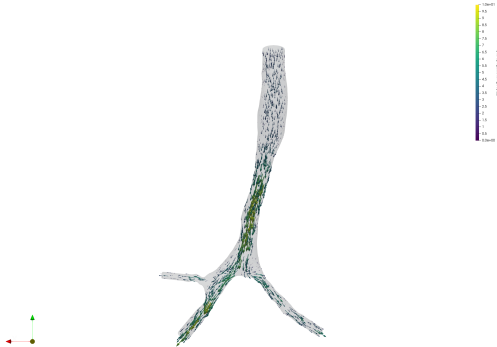
Mesh sensitivity



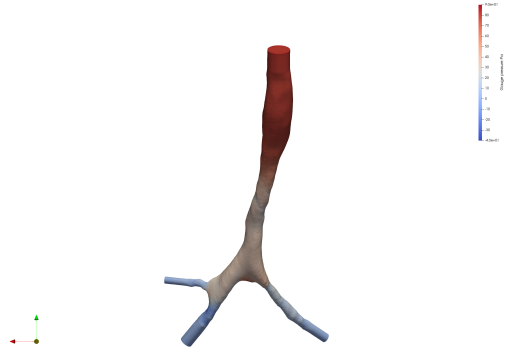
From 0.15 to 0.12 mm, selected metrics changed by at most 2.02%, while cell count increased by 92% and solver time by 157%.

Steady postoperative airflow

Velocity



Dimensional pressure



At 2 L/min: $\Delta P = 30.95$ Pa, local resistance = 15.41 Pa/(L min), peak velocity = 9.25 m/s.

Flow distribution emerges from geometry

Computed distribution

$$Q_R = 73.17\%$$

$$Q_L = 26.83\%$$

- Total inlet flow was prescribed.
- Every distal outlet had the same zero gauge pressure.
- No outlet split was imposed:

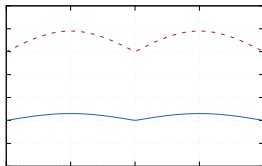
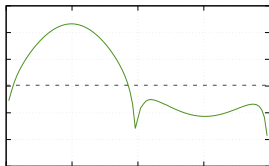
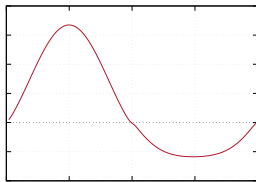
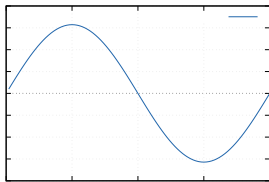
$$Q_i \sim \frac{\Delta P}{R_i}.$$

- Diameter, length, curvature, branch angle, and separation determine effective resistance.

Interpretation limit

This is geometry-driven flow division in the truncated domain, not measured physiological ventilation; peripheral resistance and compliance are absent.

Transient postoperative proof of concept



- Complete 2 s cycle
- Peak flow: ± 6.29 L/min
- ΔP : -58.7 to 167.4 Pa
- R : 5.71 – 26.63 Pa/(L/min)
- Near-zero-flow resistance masked
- 99.826% of timesteps met the PIMPLE criterion

Coarse 0.25 mm PoC, bounded upwind and $Co_{\max} = 5$: qualitative phase trends, not a timestep-independent prediction.

Flow reverses over the breathing cycle

Accelerating inspiration



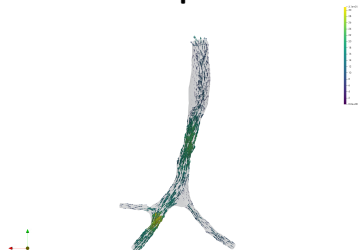
$t = 0.24$ s

Peak inspiration



$t = 0.50$ s

Peak expiration



$t = 1.50$ s

Common 0 m/s to 31 m/s colour scale and vector scale.

Conclusions and limitations

Principal findings

- Surgery increased matched area from 1.813 to 5.169 mm².
- The postoperative model produced a marked geometry-driven 73/27 right/left split.
- The 0.15 mm mesh balanced sensitivity and computational cost for steady CFD.
- The transient workflow resolves the imposed breathing phases as a PoC.

Main limitations

- Static, rigid CT geometry
- No peripheral lung impedance
- Laminar model
- No prism boundary layers
- One patient and one initial cycle
- Coarse transient fallback at high Courant number

Questions?